

# Claude Code: Operational Reference

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A no-fluff, implementation-focused breakdown. Each item lists **what / where / why**.

### 1. Core Operating Premises

- **Agent, not chatbot** — delegate tasks with a brief; let it run autonomously.
- **Verification-first** — always give Claude a way to verify its own work (tests, command output). Never accept "success" without evidence.
- **Compounding engineering** — every mistake/feedback becomes a written rule.
- **Workflow order** — Explore → Plan → Code (don't let it code first).

### 2. The .claude/ Directory Layout

| Path                                                                                                                                                                                                                                                                                                                    | Scope                 | Purpose                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------|
| .claude/                                                                                                                                                                                                                                                                                                                | Project (in repo)     | Team-shared config, committed to Git    |
| ~/.claude/                                                                                                                                                                                                                                                                                                              | Global (home)         | Personal cross-project defaults         |
| .claude/CLAUDE.md                                                                                                                                                                                                                                                                                                       | Project               | Team instructions, loaded every session |
| CLAUDE.local.md                                                                                                                                                                                                                                                                                                         | Project (git-ignored) | Personal notes/feedback                 |
| .claude/settings.json                                                                                                                                                                                                                                                                                                   | Project               | Config (permissions, env, hooks)        |
| .claude/skills/                                                                                                                                                                                                                                                                                                         | Project               | Reusable expertise units                |
| .claude/agents/                                                                                                                                                                                                                                                                                                         | Project               | Custom subagent definitions             |
| .claude/rules/                                                                                                                                                                                                                                                                                                          | Project               | Path-gated rule files (*.md)            |
| <ul style="list-style-type: none"><li>• <b>Premise:</b> Configs cascade in monorepos — nested CLAUDE.md files merge from root down.</li><li>• <b>Premise:</b> rules/*.md are path-gated (apply only to matching file paths).</li><li>• <b>Premise:</b> Prefer skills/ over legacy commands/*.md for new work.</li></ul> |                       |                                         |

### 3. CLAUDE.md (Team Guardrails)

- **Location:** .claude/CLAUDE.md (committed).
- **Purpose:** Guardrails loaded at every session start.
- **Rules for writing:**
  - Keep short — focus on preventing mistakes, not describing the codebase.
  - Let Claude write/append rules for itself after errors.
  - Avoid verbose prose.
- **Recommended sections:**
  - Development workflow (build/test/run commands)
  - Code style conventions
  - Architecture overview
  - "Gotchas" — real, recurring mistakes

### 4. CLAUDE.local.md (Personal Driver)

- **Location:** Repo root, **git-ignored**.
- **Purpose:** Capture PR feedback + personal habits to correct.
- **Operational use:**
  - Paste reviewer comments → convert into personal rules.
  - Separate "project feedback" from "personal habits."
  - Prune regularly to avoid bloat.

### 5. Skills

- **Location:** .claude/skills/<skill-name>/SKILL.md
- **Purpose:** Reusable expertise invoked via slash command.
- **Structure:**
  - Folder per skill, containing SKILL.md.
  - SKILL.md = frontmatter + instructions.
- **Capabilities:**
  - Inline shell commands via !.
  - Progressive disclosure (load detail only when needed).
- **Premise:** Check into Git → becomes institutional knowledge.
- **Use for:** Repetitive, well-defined tasks.

### 6. Custom Subagents

- **Location:** .claude/agents/<agent>.md
- **Purpose:** Isolated Claude instances for specific complex tasks (keeps main session clean).
- **Definition fields (frontmatter):**
  - name
  - description
  - tools
  - model
- **Best practices:**
  - Read-only tools for review agents.
  - Chain agents for multi-stage tasks.
  - Use git worktree for isolation.

### 7. Plugins & Marketplace

- **Command:** /plugin to browse/install.
- **Purpose:** Bundles of skills + hooks + subagents + MCP servers.
- **Premise:** Team-shared plugins speed onboarding.
- **Essential installs:**
  - /code-review
  - Language server plugins (per language).

### 8. Key Commands Reference

| Command                                                                                                                                                                                                    | Function                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| /insights                                                                                                                                                                                                  | Session analytics                     |
| /compact                                                                                                                                                                                                   | Lossy context summary (keeps session) |
| /clear                                                                                                                                                                                                     | Fresh start (drops context)           |
| /rewind                                                                                                                                                                                                    | Undo to a checkpoint                  |
| /goal                                                                                                                                                                                                      | Set verifiable completion condition   |
| /batch                                                                                                                                                                                                     | Fan out mass/parallel changes         |
| /loop                                                                                                                                                                                                      | Iterate until condition met           |
| /schedule                                                                                                                                                                                                  | Time-based task runs                  |
| /teleport                                                                                                                                                                                                  | Jump context location                 |
| /focus                                                                                                                                                                                                     | Constrain scope                       |
| /voice                                                                                                                                                                                                     | Voice input                           |
| /mcp                                                                                                                                                                                                       | List active MCP servers               |
| !                                                                                                                                                                                                          | Shell escape (run command inline)     |
| <ul style="list-style-type: none"><li>• <b>Distinction:</b> /compact = keep + summarize; /clear = wipe context.</li><li>• <b>Combo:</b> /goal + Auto mode + /focus = autonomous task completion.</li></ul> |                                       |

### 9. MCPs (Model Context Protocol)

- **What:** Standard contract letting Claude call external tools (DBs, error trackers, design canvases, notes).
- **Common servers:** GitHub, Sentry, Postgres, Obsidian.
- **Config file:** .mcp.json (project-level integration definitions).
- **Check active:** /mcp.
- **Premise — limit installs:** Too many MCPs bloat the tool list and degrade decision quality. Install only what's needed.
- **Cost note:** MCPs consume tokens (tool definitions loaded into context).

#### Obsidian Memory Tiering (if used)

| Tier | Content                         |
|------|---------------------------------|
| Hot  | Daily logs                      |
| Warm | Project notes                   |
| Cold | Decisions / long-term knowledge |

### 10. Daily Workflow Patterns

- **Bug fix:** Reproduce bug → verify reproduction → then fix.
- **Migrations / mass edits:** Use /batch.
- **Unfamiliar code:** Delegate exploration to a subagent.
- **Parallel work:** Run multiple sessions in separate git worktrees.
- **Writer/Reviewer pattern:** One session writes, a read-only session reviews.

### 11. High-Leverage Habits

- Always provide a verification method.
- Run 3–5 parallel sessions for throughput.
- Treat CLAUDE.md as a living document.
- Use /rewind instead of fighting bad state.
- Use Playwright MCP for UI verification.
- Install language server plugins for accuracy.
- Edit Claude's plan with Ctr⏏+G before execution.
- Request ASCII diagrams for architecture clarity.